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EXPERIENCE

Razorpay

Bangalore, India

Software Development Engineer Intern – Payments Platform (Settlements)

Aug 2025 – Jun 2026

- **High-Throughput Settlement Pipeline:** Engineered an async, event-driven pipeline (Golang, AWS SQS) to process transaction settlements post-hold releases. Achieved exact-once execution semantics, reliably processing **100K+ daily transactions** with automated retries and high fault tolerance.
- **Automated Reconciliation Engine:** Implemented a pull-based reconciliation cron-service to recover missed webhook status updates from downstream Payout APIs. Slashed recovery time by **99.8% (from 5 days to 15 minutes)** and eliminated **20+ monthly merchant escalations**.
- **Self-Healing Recovery Framework:** Designed a resilient recovery system to prevent settlement drops during downstream outages. Reduced manual engineering interventions (PSE tickets) by **60%** and prevented **3-hour SLA breaches** via graceful degradation and DLQ processing.
- **Fault-Tolerant CDC Pipeline:** Architected a highly available Kafka CDC pipeline for bank account synchronization across **30 partitions**. Integrated two-tier retries and DLQ isolation, ensuring **zero event loss** during network or database anomalies.
- **Microservices Migration & Observability:** Decoupled **6 core settlement APIs** from a legacy monolith into independent microservices with **zero downtime** using shadow deployments. Instrumented comprehensive telemetry (Grafana, Coralogix) across **48 API routes**, eliminating system blind spots.

TECHNICAL PROJECTS

PayFast – Distributed Event-Driven Payment Processor | Go, Kafka, Redis, PostgreSQL, Docker [GitHub](#)

- **Event-Driven Architecture:** Architected a highly available payment gateway in **Go** and **Apache Kafka**, decoupling synchronous checkout APIs from heavy background tasks to achieve **200ms p99** response times.
- **Idempotency & Double-Entry Ledger:** Eliminated duplicate charges during network retries via strict API **idempotency** (PostgreSQL locks), and engineered an ACID-compliant **double-entry ledger** to guarantee zero financial discrepancies.
- **Fraud Engine & PCI-Compliant Vault:** Designed a **sub-millisecond** Redis-backed velocity engine to intercept carding attacks, alongside an isolated Vault Service utilizing **AES-256-GCM** encryption for zero-knowledge card tokenization.
- **Settlements & Cloud Infrastructure:** Built a dynamic cron-driven engine for multi-tier batch payouts (Instant/T+1/T+24). Deployed the containerized microservices stack on **Oracle Cloud** with an Nginx reverse proxy, SSL termination, and a React dashboard.

VaultMind – Enterprise RAG & Knowledge Platform | Python, FastAPI, PGVector, Celery, Redis [GitHub](#)

- **Secure RAG & Data Isolation:** Architected a multi-tenant Knowledge Assistant using **FastAPI** and **PGVector**, enforcing strict Role-Based Access Control (RBAC) directly inside 384-dimensional **HNSW** vector queries for department-level security.
- **Async Ingestion & LLM Orchestration:** Engineered a **Celery/Redis** pipeline to offload document parsing, and built a frameworkless direct HTTP/2 integration with **Gemini 3.1** featuring exponential backoff retries for high availability.
- **Enterprise Telemetry & Sandbox:** Developed a zero-friction guest sandbox with automated 10-minute TTL cache pruning. Instrumented full **Prometheus** metrics (vector search speed, token usage) and strict SOC2-inspired audit logging.

TECHNICAL SKILLS

Languages & Frameworks: Go (Golang), Python, C++, JavaScript/TypeScript, React, FastAPI

Architecture & Backend: Microservices, Event-Driven Architecture, Distributed Systems, REST APIs, gRPC, Webhooks

Databases & Messaging: PostgreSQL, PGVector, MySQL, Redis, Apache Kafka, AWS SQS, Celery, DLQ

Cloud, DevOps & Observability: Docker, Kubernetes, AWS, CI/CD, Grafana, Coralogix, Prometheus

AI Engineering: LangChain, LangGraph, AI Agents, RAG (Vector-based & Vectorless), LLM Gateways, Guardrails, LLM Evaluation

Testing & Tooling: GoMock, Table-Driven/Integration Testing, Git, AI-Assisted Development (Cursor)

EDUCATION

Lovely Professional University

Punjab, India

B.Tech — Computer Science and Engineering; CGPA: 8.26

2022 – 2026